

“Deploying a Dockerized App on ECS with Amazon ECR”

Lab Objectives

By the end of this lab, you will be able to:

- Build a Docker image for a simple web application.
 - Push the image to **Amazon ECR**.
 - Create an **ECS Cluster** that uses **EC2 instances** (instead of Fargate).
 - Deploy and run the containerized app as a service in ECS.
 - Access the application from a web browser via the EC2 instance
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Before starting this prctl we need to first install docker in our machine top do so

Prerequisites

◆ Step 1: Download Docker Desktop

1. Go to Docker Desktop Download.
 2. Download **Docker Desktop for Windows (x86_64)**.
 3. Run the installer and follow prompts:
 - Select **WSL2 backend** (recommended over Hyper-V).
 - Allow required permissions.
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◆ Step 2: Verify Installation

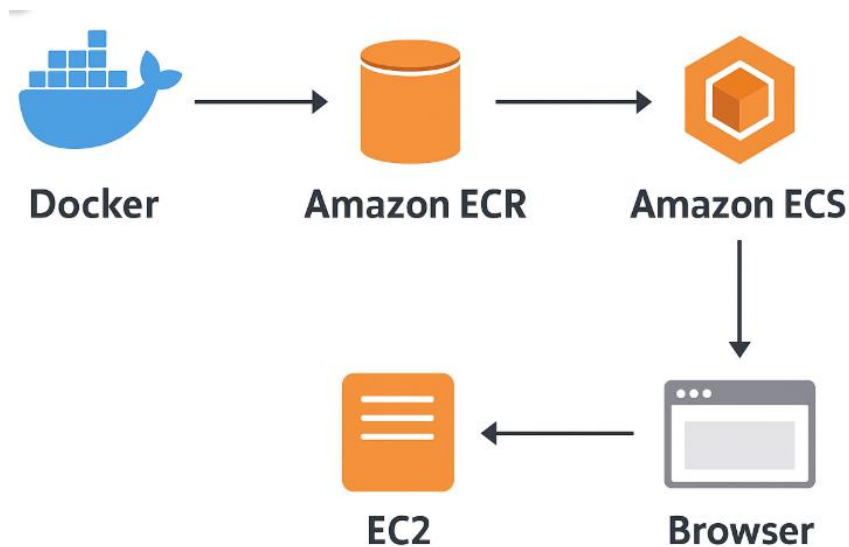
1. Open **PowerShell or Command Prompt**.
2. Run:
3. `docker --version`
4. `docker compose version`

You should see versions installed.

5. Test Docker by running:

6. docker run hello-world

If it prints a welcome message, Docker is working .



Step 1: Create ECR Repository

1. In AWS Console → **Services** → **ECR**.
2. Left menu → **Repositories** → **Create repository**.
 - **Repository name:** my-web-app
 - **Visibility settings:** Private (default).
 - Leave everything else as default.
3. Click **Create repository**.

 You'll now land on the **Repository details** page.

 At the top, you'll see a **"View push commands"** button — click it and keep those commands for later (copy to notepad).

Step 2: Push Docker Image to ECR

On your laptop/PC terminal (with Docker installed):

1. Authenticate:

```
aws ecr get-login-password --region <your-region> | docker login --username  
AWS --password-stdin <account-id>.dkr.ecr.<region>.amazonaws.com
```

2. Tag:

```
docker tag my-web-app:latest <account-  
id>.dkr.ecr.<region>.amazonaws.com/my-web-app:latest
```

3. Push:

```
docker push <account-id>.dkr.ecr.<region>.amazonaws.com/my-web-app:latest
```

✅ Refresh your repository page in the console → you should see **Image URI**.

Step 3: Create an ECS Cluster (EC2 launch type)

1. In AWS Console → **Services** → **ECS**.
2. Left menu → **Clusters** → **Create Cluster**.
3. Choose **EC2 Linux + Networking**.
4. **Cluster name:** my-ecs-cluster.
5. **EC2 instance configuration:**
 - Instance type: t2.micro (free tier).
 - Number of instances: 1.
 - Key pair: choose existing or create a new one.
 - Networking: Select an existing VPC & public subnet.
 - Security group: allow **inbound HTTP (80)** from 0.0.0.0/0.
6. Click **Create**.

✅ After a few minutes, your cluster page should show **1 registered EC2 instance**.

Step 4: Register a Task Definition

1. In ECS console → Left menu → **Task Definitions** → **Create new task definition**.
2. Select **EC2**.
3. **Task Definition Name:** my-web-task.

4. Scroll down → click **Add container**:
 - **Container name**: my-web-app
 - **Image**: paste your ECR Image URI (looks like 123456789012.dkr.ecr.us-east-1.amazonaws.com/my-web-app:latest)
 - **Memory (MiB)**: 256
 - **Port mappings**: container port 80, host port 80.
 - Click **Add**.
5. Scroll down → click **Create**.

✅ Now you'll see your new task definition listed.

Step 5: Create a Service

1. In ECS console → **Clusters** → **my-ecs-cluster** → **Services** → **Create**.
2. Launch type: **EC2**.
3. **Task definition**: select my-web-task.
4. **Service name**: my-web-service.
5. **Desired tasks**: 1.
6. **Networking**: pick the same VPC + subnet(s).
7. Click **Create service**.

✅ In a few seconds → ECS will start your task.

Check **Tasks tab** → status = RUNNING.

Step 6: Access Your App

1. Go to **EC2** → **Instances**.
2. Select your instance → copy **Public IPv4 address**.
3. Open in browser:
4. http://<EC2_PUBLIC_IP>/

🎉 You should see your app running (e.g., “Hello from ECS on EC2!”).

Step 7: Cleanup (important)

When done testing:

- Delete ECS service → then cluster.
- Delete ECR repository (or just the image).
- Terminate EC2 instance.
- Remove security group/key pair if not needed.